

3.

Be unbiased: Avoid look-ahead bias and other common testing errors

4.

Be detailed: Provide rich analytics beyond simple profit and loss

It's important to understand that backtesting isn't just about finding profitable strategies—it's about finding robust strategies that perform consistently across different market conditions and accurately reflect what would happen in live trading.

Data Preparation: The Foundation of Reliable Results

The quality of your backtest is only as good as the data it runs on. Professional algorithmic traders are obsessive about data quality—with good reason. Let's explore how to prepare data properly for backtesting.

Selecting the Right Data

The first step is gathering appropriate data for your strategy. Consider:

1.

Time period: Does your data cover different market regimes (bull markets, bear markets, ranges)?

2.

Frequency: Does the data frequency match your trading timeframe (daily for swing trading, minute data for day trading)?

3.

Completeness: Does your data include all necessary fields (OHLCV, dividends, splits)?

4.

Accuracy: Is your data free from errors and anomalies?

Let's look at how we might prepare data for backtesting:

```
1. import pandas as pd
2. import numpy as np
```